

Repository Appendix D: Detailed Results for “Benchmark-Gated Composite Mean Estimation: Regime-Conditional Improvement and Its No-Free-Lunch Limits”

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1 Relationship to the journal submission

2 This repository appendix accompanies the article and provides the longer
3 result narrative, additional result tables, and supporting visualizations. Ap-
4 pendices A–C remain in the journal supplementary file because they contain
5 the regime construction, audit evidence, and theoretical and methodological
6 details required for peer review. The version-controlled project is available
7 at https://github.com/ALRIER/Genetic_Algo_Hull_University, and
8 the journal supplement should be consulted for Supplementary Tables S1–S7
9 and Supplementary Figures S1–S3. This document does not replace the
10 journal supplementary material.

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Repository Appendix D: Detailed discovery, confirmation, audit, and external results

D.1 Initial discovery outcomes

The initial discovery stage produced a selective map of regime-dependent improvement under both discovery seeds. Under seed 101, genetic algorithm composites passed the dual benchmark gate in 9 of 18 family and regime cells; under seed 202, they passed in 7 of 18. Across both seeds, the procedure accepted composites in 16 of 36 seed-specific evaluations and retained the benchmark in the remaining 20.

The strongest family-level gains were concentrated in the Inverse Gaussian and Weibull families under both seeds. Inverse Gaussian produced three accepted cells under seed 101 and two under seed 202, with maximum $q_{0.95}$ MSE gains of 66.3% and 70.9%, respectively. Weibull produced two accepted cells under each seed, with corresponding maximum gains of 55.3% and 53.1%. Lognormal also produced two accepted cells under each seed, while Ex Wald declined from two accepted cells under seed 101 to one under seed 202. Normal and Ex Gaussian remained benchmark-dominant under both discovery seeds.

The individual gain columns in Table 1 of the main text must be read together with the dual gate. For example, Ex Gaussian produced a positive $q_{0.95}$ point estimate under seed 202, but mean MSE deteriorated and the candidate was rejected. A composite was counted as a discovery win only when both mean MSE and $q_{0.95}$ MSE improved simultaneously.

The accepted weight vectors remained directly interpretable. Depending on the regime, the composites concentrated their mass on familiar components such as the mean, median, 20% trimmed mean, geometric mean, Huber location, and Tukey biweight location. Figure D1 shows that the accepted candidates generally concentrated their weight on a limited subset of components. The discovery process therefore produced interpretable combinations of established location estimators without applying opaque transformations to the data.

D.2 Fixed-weight confirmation and seed stability

The fixed-weight evaluation was substantially more selective than the initial discovery stage. The 25 focal candidates were assessed using eight additional validation seeds, $R = 500$ Monte Carlo replicates, $B = 500$ paired

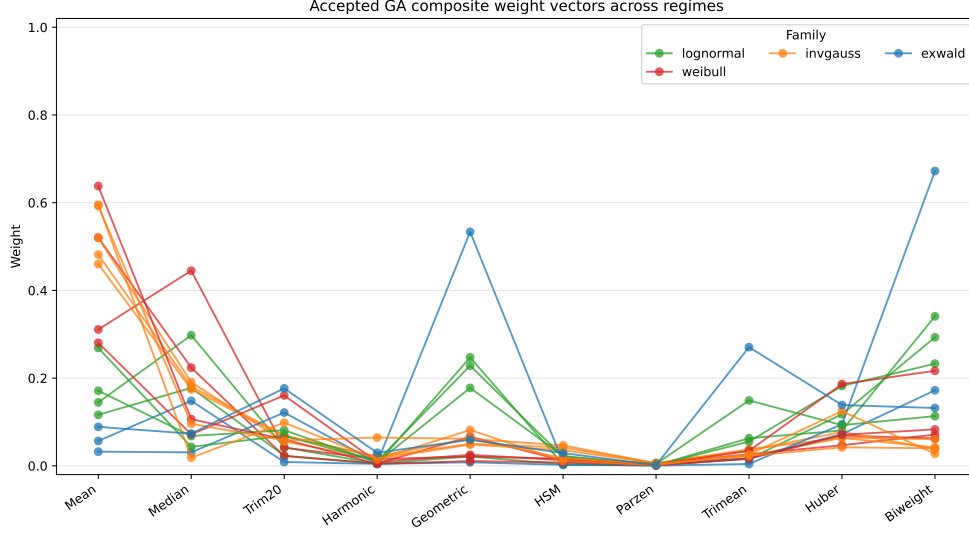


Figure D1: Accepted genetic algorithm weight vectors from the initial discovery stage. Each line represents one accepted composite estimator, and the axes correspond to the ten estimators in the initial convex basis.

bootstrap resamples, the expanded modern benchmark set, and both original and locked-unseen validation modes. Under this stronger comparison, transferable evidence was concentrated in the Lognormal family.

Three Lognormal candidates retained different levels of support (Table 2 of the main text). CV-019 and CV-010 passed the expanded gate in the locked-unseen mode under all eight validation seeds, and every seed-specific $q_{0.95}$ paired bootstrap interval was entirely above zero. CV-019 was the strongest result, with a mean $q_{0.95}$ MSE improvement of 16.3% and an envelope across the eight seed-specific intervals from 7.6% to 23.2%. CV-010 produced a smaller mean improvement of 2.2%, with a corresponding envelope from 1.6% to 3.1%. This smaller result is statistically confirmed under the study gate but operationally modest; it should not be read as an automatic recommendation to replace a simpler benchmark. Seed-specific gains, interval endpoints, and dual-gate outcomes are reported in Supplementary Table S2 in the journal supplementary material.

CV-011 differed from these two confirmed candidates. Its mean $q_{0.95}$ MSE improvement across the eight original-regime validation seeds was 3.8%. The range from 2.8% to 5.0% is the envelope obtained from the minimum lower and maximum upper endpoints of the eight seed-specific paired bootstrap

intervals. It is not a pooled confidence interval and was not calculated conditionally on the two successful seeds. All eight seeds showed a positive $q_{0.95}$ point gain, but the dual gate was cleared only under seeds 303 and 505 because mean MSE improved only under those two seeds. CV-011 did not retain its advantage in the locked-unseen mode and was interpreted as a local $q_{0.95}$ signal with no evidence of transferability. The complete eight-seed results are reported in Supplementary Table S2 in the journal supplementary material.

Figure 1 of the main text displays the seed-averaged $q_{0.95}$ gains and the envelopes of the seed-specific paired bootstrap intervals for these three signals. The envelopes for CV-019 and CV-010 remain entirely to the positive side of the zero-gain reference line, consistent with their eight-seed locked-unseen confirmation. CV-011 also has a positive $q_{0.95}$ envelope in the original regime, but this does not represent dual-gate confirmation: six seeds failed because their mean MSE gains were nonpositive.

The family-level decision matrix further demonstrates the selectivity of the expanded gate (Figure D2). Lognormal was the only family to produce a confirmed signal in the locked-unseen mode during this stage. Weibull, Inverse Gaussian, and Ex Wald, despite their initial discovery wins, reverted to the strongest modern benchmark when the original ten-component weights were frozen and evaluated under the expanded comparison.

Normal and Ex Gaussian continued to serve as negative controls. Neither family produced a locked-unseen pass, and both remained benchmark-dominant. Fixed-weight confirmation reduced the number of accepted candidates and showed that the initial discovery map did not hold under stronger validation.

The seed-level pattern provides a further distinction between transferable and local evidence. CV-019 and CV-010 passed the expanded gate under validation seeds 303, 404, 505, 606, 707, 808, 909, and 1010. Their survival across all eight seeds suggests that the confirmed result was not driven by an isolated random stream. CV-011 passed only under two seeds in the original mode, which supports its classification as a local candidate. The complete seed-by-candidate matrix is reported in Supplementary Figure S1 in the journal supplementary material.

The negative control pattern was equally stable. Normal and Ex Gaussian produced no genetic algorithm passes in the locked-unseen mode across the eight validation seeds. These abstentions show that the expanded gate did not systematically favor composites produced by the genetic algorithm. It

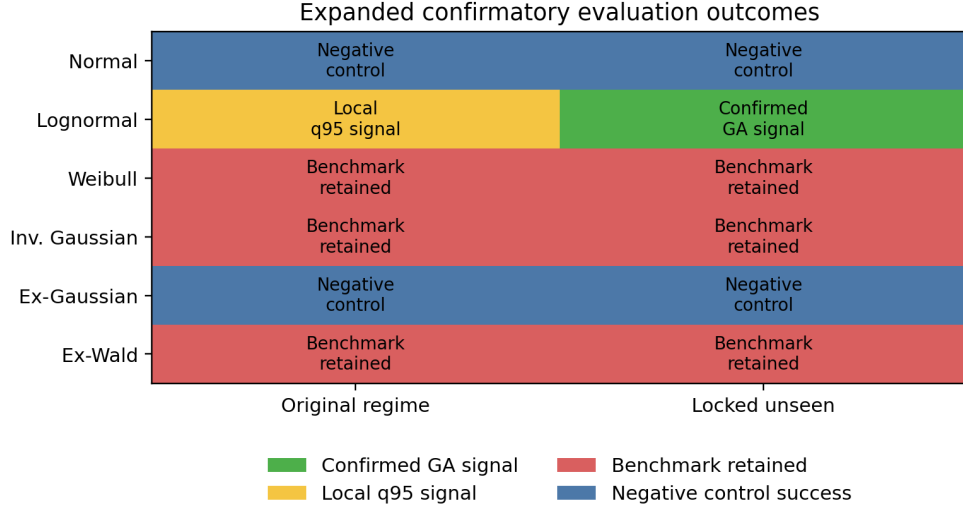


Figure D2: Fixed-weight confirmation outcomes by distributional family and validation mode. Green indicates locked-unseen confirmed genetic algorithm signal, yellow indicates a local original-regime $q_{0.95}$ signal, red indicates benchmark retention, and blue indicates negative-control success.

103 retained the benchmark when the modern estimator set was superior and
 104 accepted a composite only when its frozen weights continued to provide an
 105 advantage under stronger and previously unavailable conditions.

106 *D.3 Expanded-basis rediscovery and transfer specialists*

107 The expanded-basis search accepted 12 genetic algorithm candidates un-
 108 der the 26-component gate (Table D1). Accepted candidates emerged in five
 109 of the six distributional families: Ex Wald (3), Inverse Gaussian (4), Lognor-
 110 mal (2), Weibull (2), and Normal (1). Ex Gaussian remained benchmark-
 111 dominant. The largest discovery-stage $q_{0.95}$ MSE improvement appeared in
 112 the Inverse Gaussian family at 38.2%, followed by Weibull at 23.6% and Ex
 113 Wald at 20.6%.

114 The expanded estimator basis changed the family-level specialist map.
 115 Inverse Gaussian and Ex Wald produced the largest discovery gains, while
 116 the two Weibull candidates later provided the strongest evidence of transfer
 117 under fixed-weight validation. This result shows that introducing modern ro-
 118 bust estimators directly into the learnable simplex opened candidate regions
 119 that were unavailable to the original ten-component search.

120 The confirmed Lognormal candidates from the initial stage, CV-019 and
 121 CV-010, entered the expanded search as controlled warm-start priors. They
 122 competed under the same 26-component gate as all other candidates and re-
 123 ceived no automatic acceptance. Lognormal still produced two accepted can-
 124 didates, showing that the inherited weight structures remained competitive
 125 after Catoni-type, median-of-means, winsorized, and tuned Huber estimators
 126 became available within the search space.

127 Supplementary Table S7 in the journal supplementary material contrasts
 128 the two discovery stages, and Figure 2 of the main text shows the correspond-
 129 ing change in the family-level specialist map. The initial fixed-weight con-
 130 firmation identified its strongest transferable signals in Lognormal regimes.
 131 The expanded-basis validation added two Weibull transfer specialists, ex-
 132 tending the confirmed specialist set to a different distributional family and
 133 contamination structure.

Table D1: Expanded basis discovery outcomes by family under the 26-component genetic algorithm basis. Gains are discovery stage relative improvements before fixed-weight validation. Transfer specialist status was assigned only after the subsequent post-discovery fixed-weight validation.

Family	Expanded stage GA wins	Confirmed prior inherited	Best $q_{0.95}$ gain	Source regimes
Normal	1	No	7.0%	REG02/s202
Lognormal	2	Yes: CV-019, CV-010	3.9%	REG01/s101, REG03/s101
Weibull	2	No	23.6%	REG01/s101, REG02/s202
Inverse Gaussian	4	No	38.2%	REG01/s101, REG02/s101, REG02/s202, REG03/s202
Ex Gaussian	0	No	—	benchmark retained
Ex Wald	3	No	20.6%	REG01/s101, REG03/s101, REG03/s202
Total	12	2 priors	—	—

134 The expanded discovery universe contained 36 seed-specific family and
135 regime cells: six families, three regime classes, and two discovery seeds.
136 Twelve cells cleared the discovery gate and entered the fixed-weight panel
137 as accepted GA winners. From the 24 cells that did not clear that gate,
138 seven borderline near-gate cells and six benchmark-control cells were also se-
139 lected for diagnostic fixed-weight evaluation. The resulting validation panel
140 therefore contained 25 candidates: 12 discovery winners and 13 deliberately
141 retained nonwinners. The remaining 11 benchmark-retained discovery cells
142 were not included in this panel. The subsequent fixed-weight validation orga-
143 nized these 25 candidates into four evidence grades (Table D2). The taxon-
144 omy preserves the distinction between discovery success, diagnostic follow-up,
145 and confirmed transfer.

146 Two Weibull candidates reached transfer specialist status after producing
147 advantages supported by confidence intervals under locked-unseen conditions.
148 Ten additional candidates retained discovery-supported local evidence: three
149 Ex Wald, four Inverse Gaussian, two Lognormal, and one Normal candidate.
150 These candidates identify valid discovery regions that remain relevant for
151 further study, even though they did not meet the complete transfer standard.

152 Seven candidates were classified as near-gate cases. Their results ap-
153 proached the acceptance boundary and identify additional regions of the
154 simplex that may benefit from greater discovery budgets or more targeted
155 validation. Six Normal and Ex Gaussian cases served as negative-control
156 successes, verifying that the expanded procedure continued to retain the
157 benchmark when the composite did not provide sufficient improvement.

158 Figure 2 of the main text shows the basis-dependent change in the val-
159 idated specialist map: the initial confirmation retained Lognormal signals,
160 while the expanded basis added Weibull transfer specialists.

161 The four evidence grades were distributed selectively across families. In-
162 verse Gaussian and Ex Wald contributed several discovery-supported candi-
163 dates, confirming that the expanded basis found promising local regions in
164 both families. Weibull produced the only two transfer specialists from this
165 stage. Normal and Ex Gaussian also provided important control evidence by
166 showing that basis expansion did not lead to indiscriminate composite accep-
167 tance. The complete family-level visualization is reported in Supplementary
168 Figure S2 in the journal supplementary material. Candidate-level discov-
169 ery gains, frozen locked-unseen gains, original-regime gains, and seed-level
170 CI support for all 25 evaluated candidates are reported in Supplementary
171 Table S3 in the journal supplementary material.

Table D2: Evidence taxonomy from the 25-candidate post-discovery fixed-weight validation panel ($N_{\text{EST}} = 26$). The panel contains 12 accepted discovery winners, seven borderline near-gate cells, and six benchmark-control cells. The grades summarize the strength and scope of the available evidence without changing the discovery or validation gate decisions.

Evidence grade	N	Families and interpretation
transfer specialist	2	Weibull; confidence-interval-supported locked-unseen advantage
discovery-supported local candidate	10	Ex Wald (3), Inverse Gaussian (4), Lognormal (2), Normal (1)
near-gate candidate	7	Ex Gaussian (3), Normal (2), Weibull (2); approached the validation gate
negative-control success	6	Normal (3), Ex Gaussian (3); benchmark correctly retained

172 The two Weibull transfer specialists are reported in Table 3 of the main
173 text. Q1R011, discovered in Weibull REG01 under seed 101, improved mean
174 MSE by 2.79% and $q_{0.95}$ MSE by 2.72% under locked-unseen validation.
175 Q1R012, discovered in Weibull REG02 under seed 202, achieved correspond-
176 ing gains of 3.18% and 1.25%.

177 The contraction from discovery to frozen locked-unseen validation was
178 substantial and must be evaluated separately for each candidate. Q1R011’s
179 $q_{0.95}$ gain declined from 7.32% at discovery to 2.72%, a reduction of 62.8%.
180 Q1R012’s $q_{0.95}$ gain declined from 23.61% to 1.25%, a reduction of 94.7%.
181 The 23.6% family maximum in Table D1 belongs to Q1R012 and should not
182 be used as the discovery value for Q1R011. These reductions support the
183 separation of search and frozen validation and are consistent with substantial
184 discovery-stage optimism. Because locked-unseen evaluation also changes
185 the contamination profile, the reductions do not isolate a pure winner’s-
186 curse effect. The corresponding stage-specific quantities for every evidence
187 category are shown in Supplementary Table S3 in the journal supplementary
188 material.

189 Across the eight locked-unseen validation seeds, the criterion-specific com-
190 parators for these two specialists were Huber, tuned Huber, or winsorized
191 estimators, with no Catoni-type variant serving as the comparator. Their
192 confirmed gains therefore did not result from comparison with an unusually
193 weak Catoni-type tuning.

194 These gains were obtained against the strongest expanded benchmark
 195 after the candidate weights had been frozen. Both specialists also satisfied the
 196 validation requirements across all eight additional seeds. Their performance
 197 indicates that the expanded search identified reproducible weight structures
 198 for severe Weibull contamination conditions, including upper-tail and point-
 199 mass contamination at scale 30.

200 The specialist advantage was specific to the locked-unseen conditions. Un-
 201 der the original-regime evaluation, Q1R011 recorded changes of -15.77% in
 202 mean MSE and -45.06% in $q_{0.95}$ MSE. Q1R012 recorded changes of -15.20%
 203 and -34.56% , respectively. These results define the applicability boundary
 204 of the two estimators: their frozen weights improve performance in the val-
 205 idated extreme transfer conditions, but their use requires regime matching
 206 and should not be extended automatically across all Weibull settings.

207 Figure D3 displays the contrast between the two validation modes. Both
 208 locked-unseen $q_{0.95}$ estimates are positive, showing a reproducible advantage
 209 under the severe transfer conditions. The negative original-regime estimates
 210 mark the boundary outside which the specialists should not be activated.

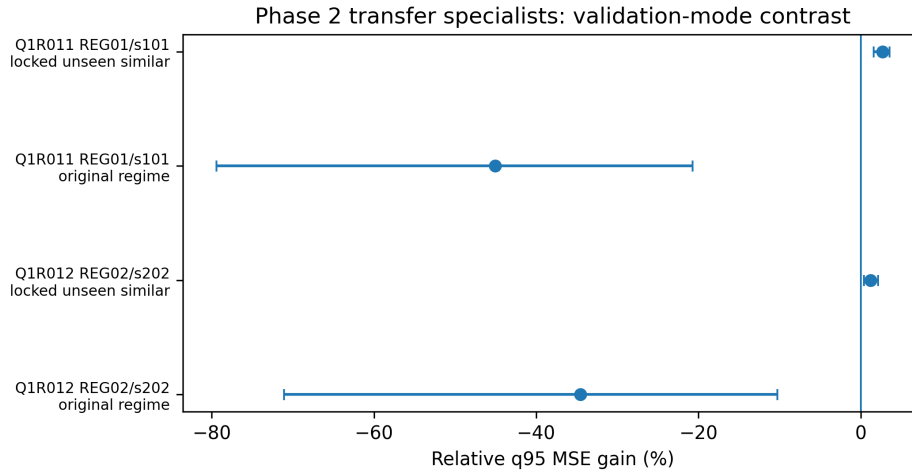


Figure D3: Validation mode contrast for the two Weibull transfer specialists. Points show $q_{0.95}$ MSE gains averaged across validation seeds, and intervals show the seed-level bootstrap confidence interval envelope across the eight validation seeds. Both specialists achieve positive gains under locked-unseen validation, while the original-regime results define the limits of their validated application.

211 *D.4 Confirmation and abstention audits*

212 The confirmation and abstention audits described in Section 3.4 of the
 213 main text preserved both Weibull specialist confirmations. No additional
 214 candidate met the complete confirmation standard.

215 The robustness audit showed that both specialists substantially reduced
 216 the influence of extreme observations. One arbitrarily large observation
 217 moved Q1R011 by approximately 1/62 of its effect on the sample mean and
 218 moved Q1R012 by approximately 1/46. Most of the frozen weight in both
 219 composites was assigned to procedures with bounded or attenuated influence,
 220 including Huber, Catoni-type, biweight, median, and winsorized estimators.
 221 The geometric and harmonic means together accounted for no more than
 222 2.3% of the total weight.

223 Both specialists retained a small positive weight on the sample mean,
 224 with

$$w_{\text{mean}} = 0.016$$

225 for Q1R011 and

$$w_{\text{mean}} = 0.022$$

226 for Q1R012. Their formal breakdown point consequently remains zero. This
 227 theoretical limitation does not remove the observed reduction in gross-error
 228 sensitivity, which was one to two orders of magnitude below that of the
 229 sample mean. The corresponding frozen-weight audit is reported in Supple-
 230 mentary Table S4 in the journal supplementary material.

231 The bootstrap audit found that modal components contributed very lit-
 232 tle to either specialist. Their combined modal weights were 0.74% for the
 233 Weibull REG01 specialist and 0.43% for the Weibull REG02 specialist. The
 234 known limitations of the ordinary bootstrap for modal estimators are con-
 235 sequently unlikely to affect the confirmed intervals materially (Ghosh et al.,
 236 1984; Romano, 1988).

237 Median weight was more visible in selected candidates. It reached 17.9%
 238 in the Weibull REG01 specialist and 13.5% in the Lognormal REG01 can-
 239 didate. These values justify continued caution when interpreting intervals
 240 containing a substantial median contribution. In both cases, however, the
 241 median formed part of a broader weighted estimator whose remaining mass
 242 was concentrated largely on smooth components with $\sqrt{n}$ consistency.

243 Multiplicity correction reinforced the stability of the specialist results.
 244 Both Weibull candidates survived the Benjamini–Hochberg correction at the
 245 5% level across all eight validation seeds, and their $q_{0.95}$ lower confidence

246 bounds remained strictly above zero. The aggregate one-sided signal was at
247 or below the bootstrap resolution,

$$1/B = 1/500$$

248 per seed. The correction preserved the two confirmed specialists and did not
249 elevate any additional candidate.

250 The split-sample audit closely reproduced the pooled estimates. Q1R011
251 achieved a held-out $q_{0.95}$ MSE gain of 2.84%, compared with its pooled gain of
252 2.72%. Q1R012 achieved a held-out gain of 1.28%, compared with a pooled
253 gain of 1.25%. The differences were only 0.12 and 0.03 percentage points,
254 respectively, providing additional evidence that the reported gains were not
255 driven by selection within the complete validation seed collection.

256 The validation seeds 303 through 1010 were distinct from the discovery
257 seeds 101 and 202, adding an independent layer to confirmation. The earlier
258 CV-019 Lognormal specialist remains the largest selected result on a struc-
259 turally similar locked-unseen regime, with a $q_{0.95}$ MSE gain of 16.3% and a
260 lower confidence bound of 7.6%. Because each half of the split-sample audit
261 contains only four seeds, those estimates are used as robustness checks and
262 not as separate inferential intervals.

263 The random Dirichlet audit also supported the selectivity of the bench-
264 mark gate. Among the 34 regime and validation mode cells, 23 confirmed
265 abstention because none of the 4,000 sampled composites cleared the dual
266 gate with stability across all eight seeds. Nine of the 17 audited regimes
267 confirmed benchmark retention in both validation modes.

268 The audit successfully distinguished positive and negative controls. All
269 six Normal and Ex Gaussian control regimes confirmed abstention; no ran-
270 dom vector passed the gate across all eight seeds. The two frozen Weibull
271 transfer specialists each passed under all eight seeds when included as pos-
272 itive controls. This contrast shows that the audit could recover established
273 specialist signals while continuing to retain the benchmark in the designated
274 control families. Complete audit counts are reported in Supplementary Ta-
275 ble S5 in the journal supplementary material.

276 The Dirichlet audit also identified new configurations for future investi-
277 gation. Eight heavy-tailed regimes contained at least one random composite
278 that passed across all eight seeds but had not been recovered by the com-
279 pleted genetic algorithm run. These signals occurred among Ex Wald and
280 Inverse Gaussian regimes, two Lognormal regimes, and one Weibull near-gate

281 regime. No corresponding signal appeared in the Normal or Ex Gaussian
282 control families.

283 The strongest new search target was Q1R003 in Ex Wald, where 72 of
284 the 4,000 random draws passed the stability requirement, corresponding to
285 a 1.8% pass rate. Pass rates in the remaining flagged regimes were below
286 0.3%. These results show that gate-passing configurations can be uncommon
287 under the selected Dirichlet proposal distribution and provide concrete tar-
288 gets for future persistence and frozen validation. They were not added to the
289 confirmed set because the sampled weights had not yet completed the full
290 validation procedure.

291 Taken together, the audits strengthened the main findings of the ex-
292 panded stage. They preserved both Weibull transfer specialists, confirmed
293 benchmark retention across the Normal and Ex Gaussian controls, repro-
294 duced the specialist gains under multiplicity and split-sample checks, and
295 identified eight focused search targets for future studies. The resulting evi-
296 dence supports two validated specialists while also providing a clear map for
297 extending the discovery process.

298 *D.5 External validation on public data*

299 The external battery requested 264 public registry entries, successfully
300 loaded 228, and selected 128 after balancing profile quotas and source caps.
301 Eight selected entries were not scored because their operational profiles had
302 no eligible specialist mapping: four were classified as zero-inflated/count and
303 four as empirical-unknown. The remaining 120 entries produced dataset-
304 level evaluations. Among them, 43 independent parent datasets satisfied the
305 profile eligibility requirements for primary testing.

306 Twelve of the 25 frozen candidates achieved at least one strict external
307 win. After repeated evaluations were collapsed to their independent parent
308 datasets, the candidates produced wins confirmed by confidence intervals and
309 surviving false discovery rate control on 26 of the 43 eligible parents. The ex-
310 tent of external transfer was measured across these 26 independent datasets,
311 with lower-level comparisons excluded from the independent dataset count.

312 When strict wins were summarized by empirical profile, 14 parent datasets
313 contributed to Lognormal, 12 to Normal, and 5 to Weibull. These profile-
314 specific counts are not mutually exclusive because distinct marginals and
315 depth probes from the same parent dataset could activate under different
316 empirical profiles. They consequently do not sum to the 26 unique parent
317 datasets used as the primary measure of independent dataset coverage. No

318 profile-eligible external win was observed in the Ex Gaussian, Ex Wald, or
319 Inverse Gaussian families.

320 Evidence within the successful parent datasets was also extensive. Across
321 eligible subregime comparisons, 256 rows reached the strong-win threshold
322 supported by confidence intervals. Of these, 255 survived the Benjamini–
323 Hochberg correction at the 5% level and 257 survived at the 10% level. The
324 additional two rows at the 10% threshold included one strict strong win
325 and one sensitivity-only row. Among the 255 confirmations at 5%, 203 were
326 within-dataset depth probes and were not counted as additional independent
327 datasets.

328 Figure 3 of the main text separates these two evidential dimensions. The
329 left panel reports profile-specific parent dataset coverage, and the right panel
330 reports corrected subregime confirmations within those datasets. The exter-
331 nal signal was concentrated in the same three empirical profile families in
332 which the frozen candidates activated, while the remaining families retained
333 their corresponding benchmarks.

334 The primary gate remained highly selective. The audit-ready gate table
335 contained 5,900 condition-level rows spanning multiple contamination condi-
336 tions within each dataset, specialist, and validation-mode combination. Of
337 these lower-level rows, 2,211 had a positive paired bootstrap interval and
338 were labelled strong-win cells. They are repeated condition-level observa-
339 tions, not bootstrap replicates and not independent evidence. Aggregation
340 of these rows produced 809 profile-eligible dataset, specialist, and mode com-
341 parisons. Among the 809 aggregated comparisons, 256 contained at least one
342 strong win supported by a confidence interval. After multiplicity correction,
343 255 survived at a 5% false discovery rate and 257 survived at the 10% sen-
344 sitivity threshold. The external conclusion did not depend on adopting the
345 more permissive threshold; the complete gate funnel is reported in Supple-
346 mentary Figure S3 in the journal supplementary material.

347 External survivors were observed in both the original and locked-unseen
348 related validation modes for the Lognormal, Normal, and Weibull profiles.
349 In each of these families, the lower confidence bound on the reported gain
350 remained strictly above zero. The external signal was not confined to a single
351 validation mode within the successful families.

352 Table 4 of the main text summarizes the principal external counts and
353 distinguishes independent dataset-level evidence from repeated confirmations
354 within those datasets.

355 These results reinforce the interpretation of conditional transfer and

356 match the pattern observed in the simulation analyses. The frozen candi-
357 dates improved on the prespecified profile benchmarks in 26 of the 43 eligible
358 independent parent datasets. These activation areas also produced 255
359 corrected confirmations across profile-eligible subregime comparisons. The
360 first result establishes transfer across independent empirical datasets; the
361 second shows repeated support within the profiles where activation occurred.

362 This selective pattern also aligns with the No Free Lunch framing devel-
363 oped throughout the article. Broad dominance across most public datasets
364 would not fit a regime-dependent specialist interpretation. The external gains
365 remained concentrated in empirical profiles aligned with the validated spe-
366 cialist configurations. When the empirical profile did not support candidate
367 activation, the framework retained the benchmark.

368 The external evidence remains limited to the operational target and
369 contamination definition used in the battery. Both were constructed from
370 each dataset’s own marginal distribution. The findings establish profile-
371 conditional recovery of the full-sample mean from repeated subsamples, but
372 they do not validate performance against an independently known popula-
373 tion mean. Within this scope, the external battery reinforces the simulation
374 findings: the frozen composites activate selectively, reproduce their advan-
375 tage under matched empirical conditions, and abstain outside the profiles
376 supported by the evidence.

377 **References**

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